Nuclear Fuel Performance

NE-533

Spring 2026

Housekeeping

- Grading of exam 2 and MOOSE project is still ongoing…
- Lecture 13 was uploaded last night/this morning
- Today is Lecture 14
- Lecture 15 will be an online makeup
- Then lecture 16 then problem session + Exam 3
- Exam 3 should be March 31

- Have a good spring break next week!

Paper Project

- Will upload papers with names attached today
- Will deliver a 15-minute presentation providing a critical review of the paper
- This is basically a mini QE Part 1, without the report
- If you have a topic that you want the paper to address, reach out to me ASAP
- Upload slides to moodle by April 3, will have an assignment generated
- Class presentations will be conducted in-person on April 7 and 9

  we get through as many as we can, and the remaining students will be tasked with uploading a recording

  distance students will upload a recording
- Not assigned for NE433 students

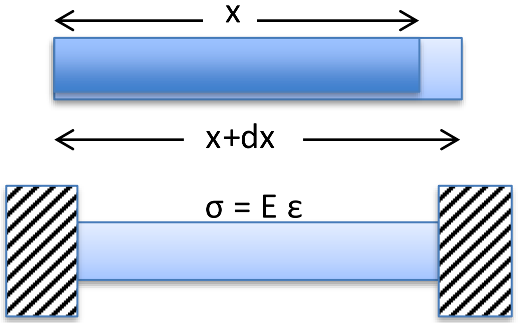
Last Time

- Fission gas release
  - experiments fall into post-irradiation annealing, or in-pile
- Showed Booth model equations for post-irradiation annealing and in-pile release
- Forsberg-Massih 2-stage FGR model
- Three components of diffusion
  - intrinsic, RED, RDD
- Swelling/dimensional change
  - densification, thermal expansion, solid/gas fission product swelling, creep

FUEL SWELLING/DIMENSIONAL CHANGE CONT…

Creep

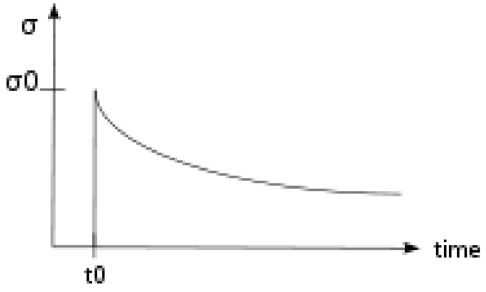
- Creep is a general mechanism for plastic deformation that occurs over time when $<\ _{y}$
- Consider a heated metal beam so it expands some distance dx



- We then fix it between two walls and let it cool down
- Because $<\ _{y}$, that stress remains constant
- In creep, defect diffusion is induced by the stress to cause permanent deformation and reduce the stress
- Therefore, creep

  Occurs over time

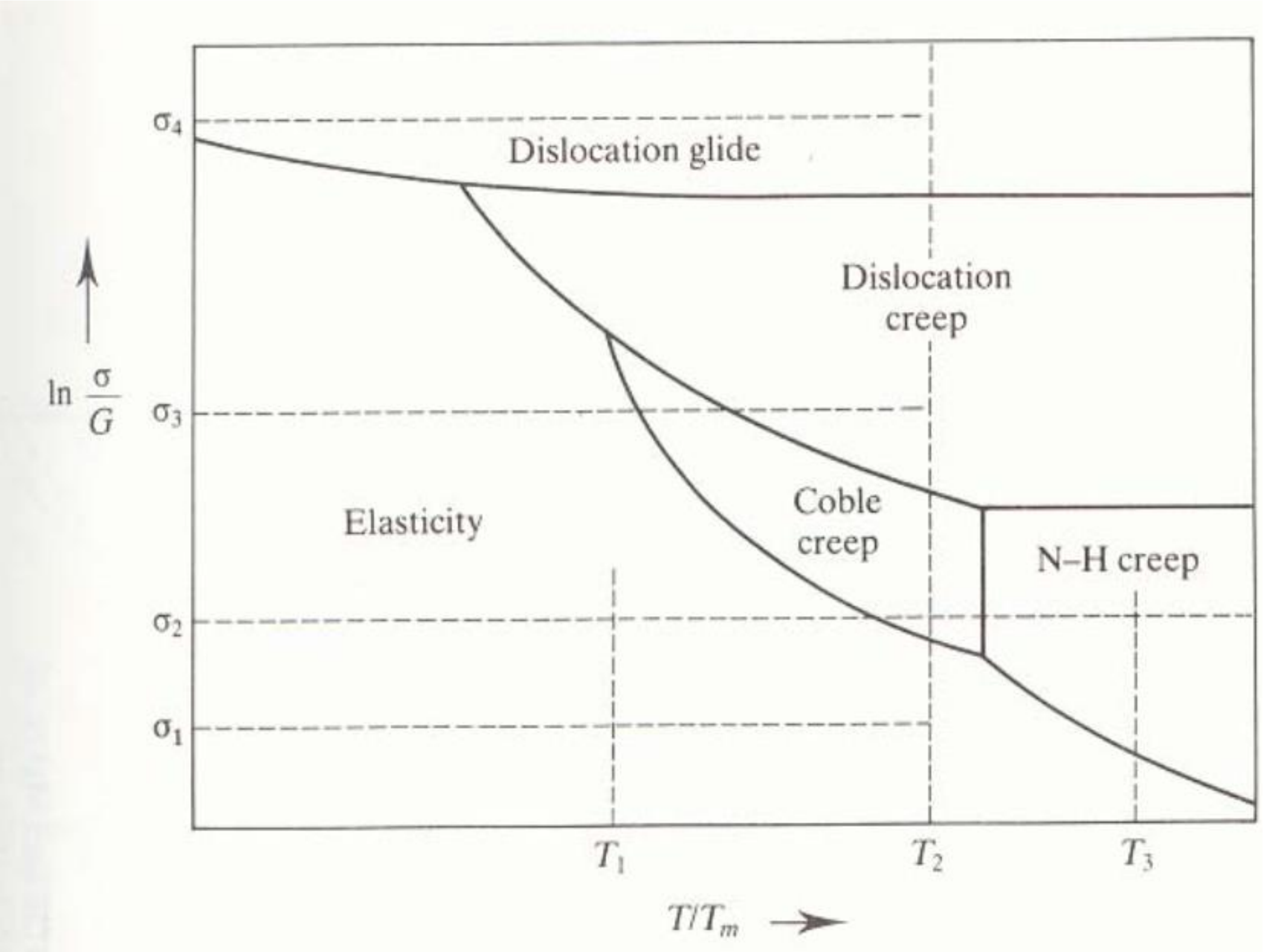
  Increases with increasing number of diffusing defects
  - High temperature (**thermal creep**)
  - Irradiation (**irradiation enhanced thermal creep**)



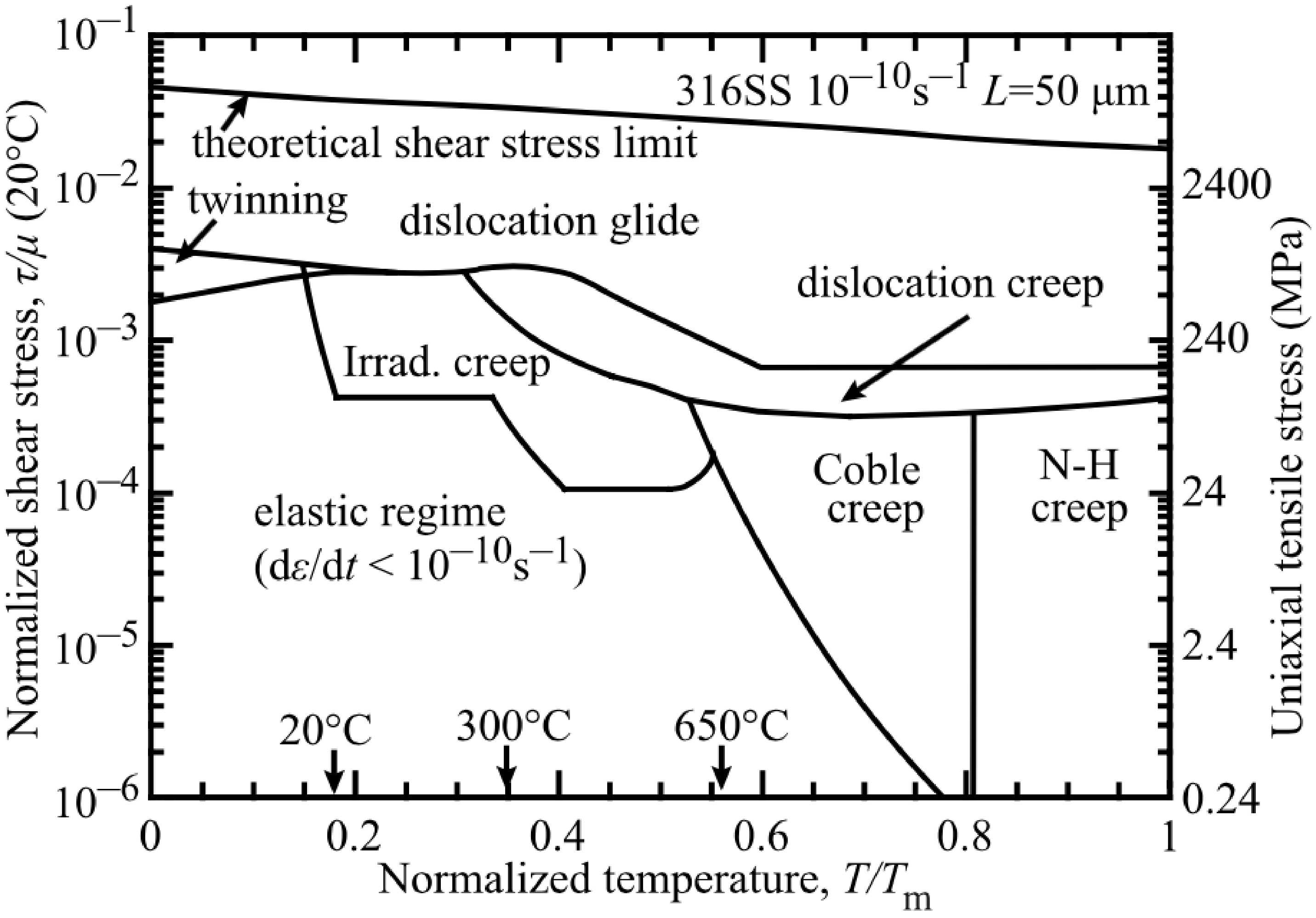
Creep

- General creep equation: $$\dot{\epsilon} = \frac{C\sigma^m}{D_{gr}^b} e^{\frac{-Q}{k_b T}}$$
- Creep can be caused by various microstructural mechanisms
- Bulk Diffusion (Nabarro-Herring creep)
  - Atoms diffuse (high T), causing grains to elongate along the stress axis
  - Q = Q(self diffusion), m = 1, and b = 2
- Grain boundary diffusion (Coble creep)
  - Atoms diffuse along grain boundaries to elongate the grains along the stress axis
  - Q = Q(grain boundary diffusion), m = 1, and b = 3
- Dislocation creep (power law creep)
  - Dislocations glide under a high stress
  - Dislocations climb due to defects to avoid obstacles
  - Q = Q(self diffusion), m = 4 6, and b = 0

Different creep mechanisms are active for different combinations of stress and temperature



The behavior of creep changes in irradiated materials



Irradiation and Creep

- Irradiation accelerates creep, causing it to be significant at lower temperatures
- Irradiation typically has little effect on diffusional creep (but can increase defect concentrations)
- Primary impact is accelerating dislocation creep in structural/cladding materials
- The dislocation creep rate can be written as $\dot{\varepsilon} = \rho_d^m b \mathrm{v}_d$

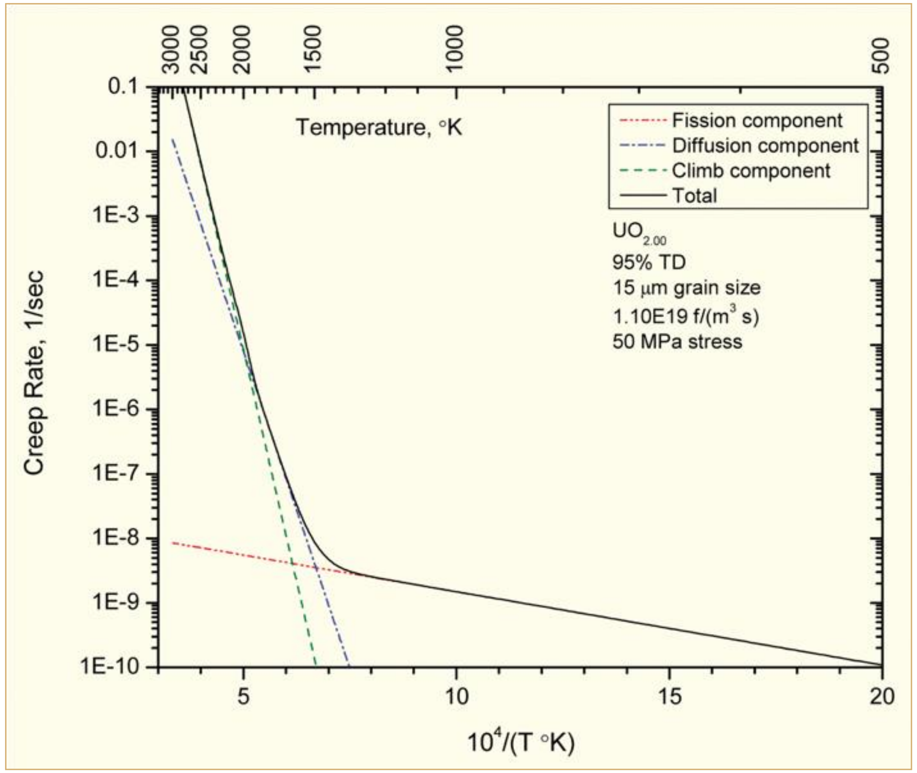
  ρ_d^m is the density of mobile dislocations

  b is the burgers vector

  v_d is the dislocation velocity
- Gliding dislocations quickly get pinned by obstacles
- As the dislocations absorb defects created by irradiation, they climb to different slip planes to avoid the obstacles
- More interstitials are absorbed than vacancies due to the higher sink strength for interstitials

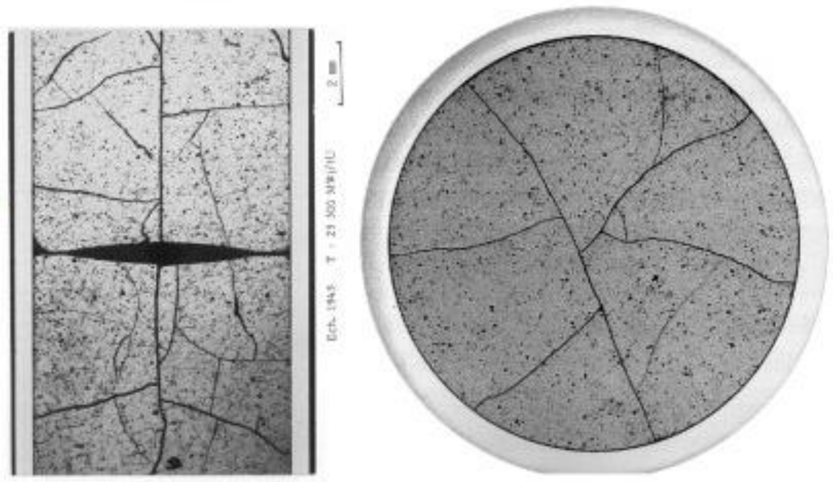
Fuel Creep

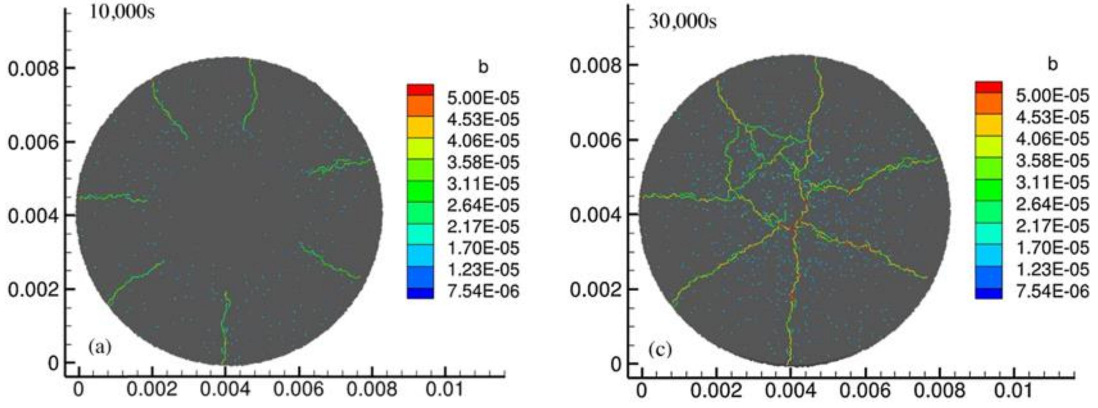
- Like other materials, the fuel also undergoes creep
- The fuel creep (In UO2) is a combination of diffusion creep and irradiation creep
- Right shows the combined creep rate
- It is expected that fuel creep plays a major role in dimensional change in metallic fuels, largely via N-H and Coble creep, but still unproven experimentally and no good creep models exist for metallic fuels



Fracture

- UO_2 pellets fracture during changes in temperature due to large thermal stresses
- Fracture results in:
    - Gap reduction
    - Reduced thermal conductivity
    - Increased avenues for fission gas release
- Fracture has been typically modeled in two ways:
    - Empirical relocation model that is a function of burnup
    - Semi-empirical smeared cracking model
- Modern methods provide means of modeling discrete cracks





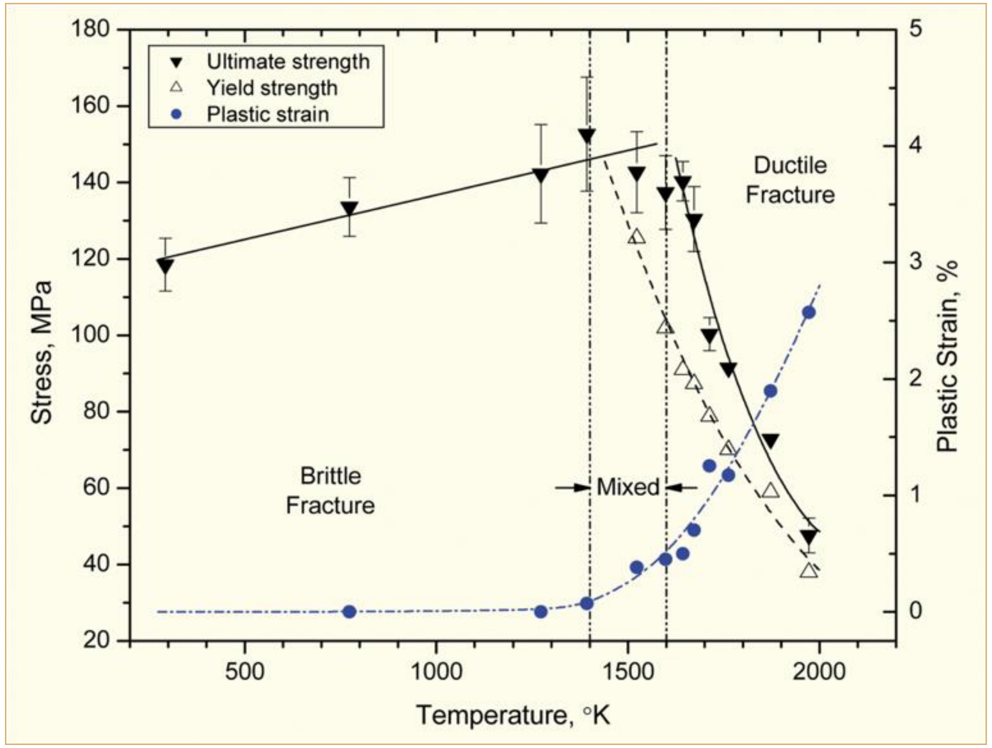
- Radial cracks partially penetrate the pellet during temperature increase
- Full cracking occurs when the temperature decreases

Fracture

- The fracture behavior of the fuel is fairly complicated
- Fracture strength varies with grain size (G)

  $\sigma_{frac} = G^{-m}\sigma_{frac,\ ref}$, m = 0.04   0.05 (vs. m ~ 0.5 for metal)

  Increasing grain size from 10 μm to 100 μm reduces σ_{frac} by ~10%
- Ductility transition temperature is lower in-reactor than in thermal tests
- Fracture strength is ~10 × higher in compression than in tension
- Load-deformation behavior strongly affected by creep under in-reactor conditions



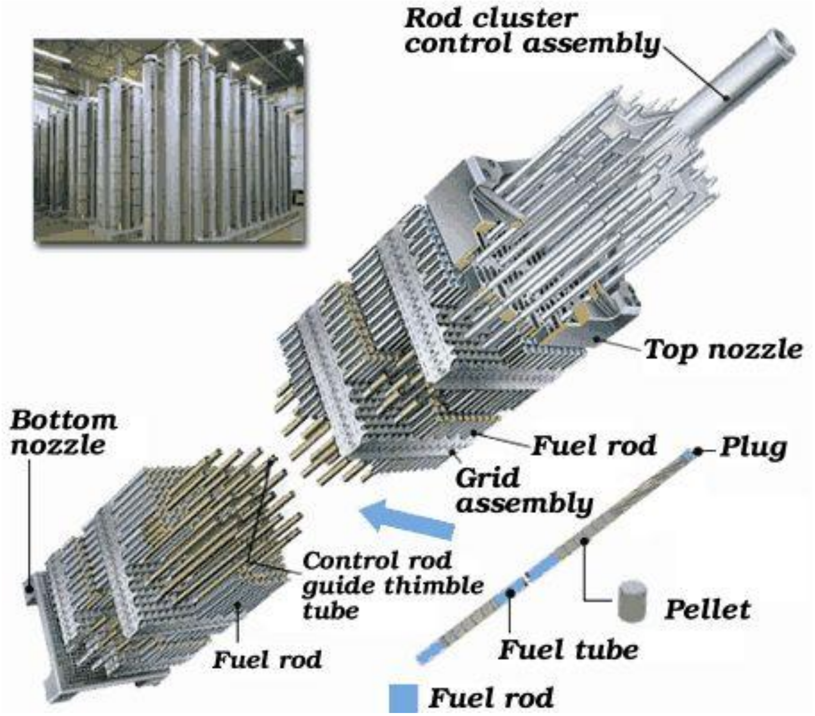
Summary

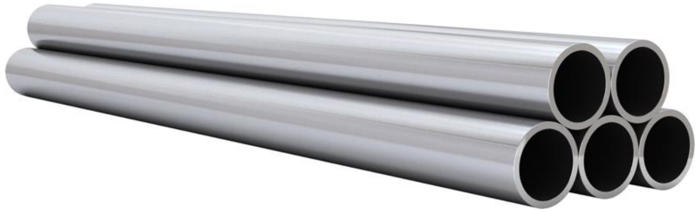
- Many materials models for fuel are empirical and correlated to burnup
- Fuel pellets change shape due to
    - Thermal expansion (increase in volume)
    - Densification (decrease in volume)
    - Swelling (increase in volume)
    - Creep (volume stays the same)
- Fracture also decreases the gap, as fractures pieces shift outward

ZIRCONIUM CLADDING

Cladding

- The purpose of the cladding is to:
    - Hold the pellets together so that coolant can freely flow past
    - Transport heat from fuel to the coolant
    - Contain fission products
    - Contain fuel fragments
- We would also prefer if the cladding had little to no impact on the neutron transport in the reactor





Why Zirconium alloys?

Benefits

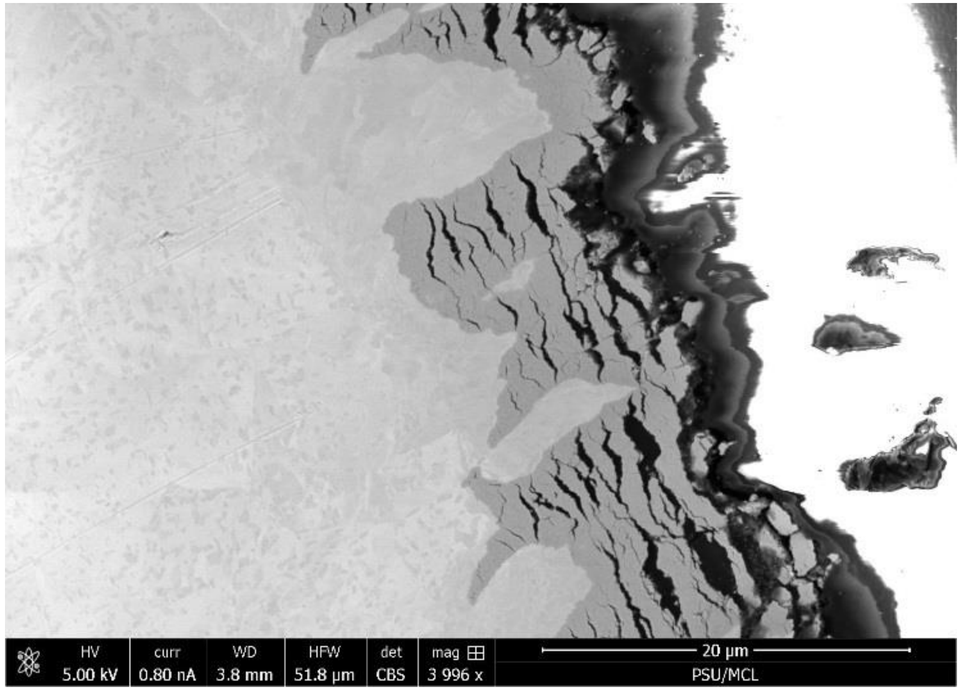
- Low neutron cross section
- Corrosion resistance in 300 C water
- Resistance to void swelling
- Adequate mechanical properties
- Good thermal conductivity
- Affordable cost
- Available in large quantities

Problems

- Corrosion under high-temperature steam
- Hydride embrittlement
- Anisotropic characteristics lead to creep and growth

Zirconium

- Pure zirconium was not acceptable due to its oxidation behavior
- Oxygen with water reacted to form an oxide layer
- The oxide layer is brittle and flakes off, allowing more oxidation to occur
- Zr alloys were developed to improve corrosion resistance



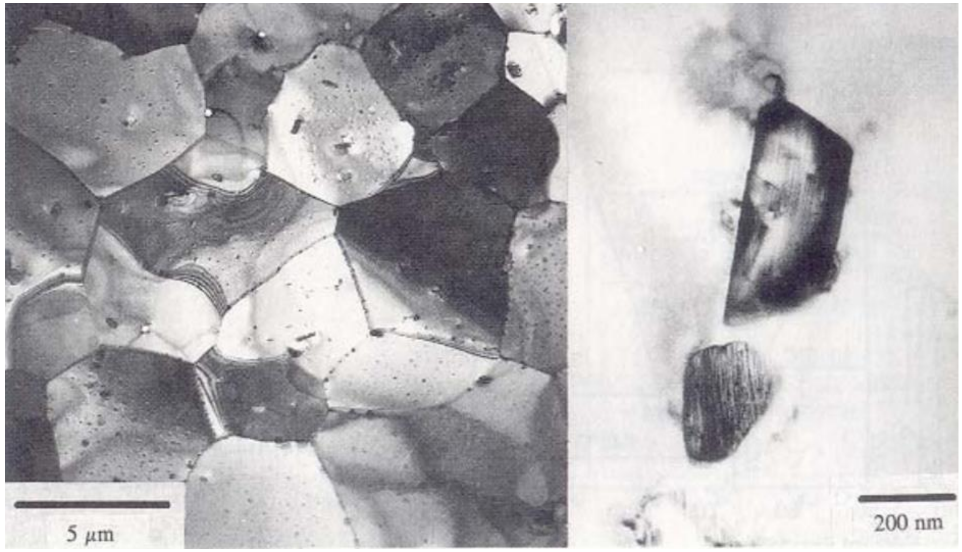
Commercial Zr Alloys in PWRs

Alloy	Sn %	Nb %	Fe %	Cr %	Ni %	O %
PWRs (structural components and fuel rods)						
Zircaloy-4 (SRA)	1.2-1.7	-	0.18-0.24	0.07-0.13	-	0.1-0.14
ZIRLO (SRA)	1	1	0.1	-	-	0.12
Optimized ZIRLO (pRXA)	0.7	1	0.1	-	-	0.12
M5 (RXA)	-	0.8-1.2	0.015-0.06	-	-	0.09-0.12
HPA-45 (SRA/RXA)	0.6	-	Fe+V	-	-	0.12
NDA (SRA)	1	0.1	0.3	0.2		0.12
MDA (SRA)	0.8	0.5	0.2	0.1		0.12

PRXA Partial Recrystallization Anneal
RXA Recrystallization Anneal
SRA Stress-Relief Anneal

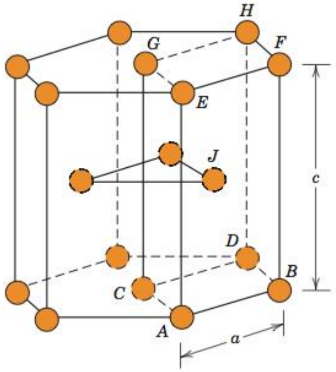
Alloying Elements

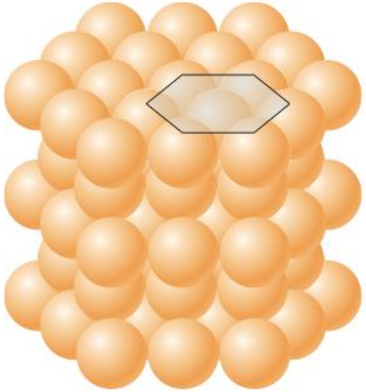
- The alloying elements impact the microstructure of the material
- The intermetallic precipitates significantly improve the oxidation behavior
- In Zircaloy 2, the precipitates are
  - $Zr(Cr, Fe)_2$
  - $Zr_2(Ni, Fe)$
- In Zircaloy 4, the precipitates are
  - $Zr(Cr, Fe)_2$
- Phosphides (Zr_3P) and silicides (Zr_3Si) are also occasionally found
- The precipitate distribution, size, morphology, and composition all impact the properties of the material

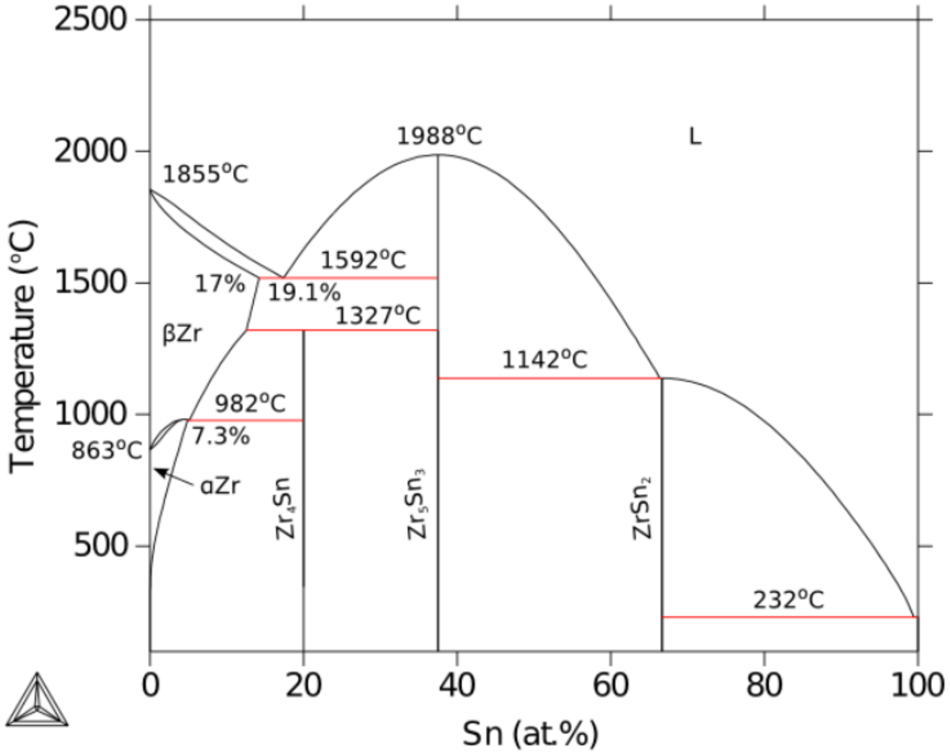


Zirconium Phases

- The -Zr phase has a hexagonal-close-packed (HCP) structure
    - At temperatures below about 863ºC
    - Has the most desirable properties
- The -Zr phase has a body-centered cubic (BCC) structure
    - We try to avoid this phase







Zr Tube Fabrication

- The cladding tubes are fabricated using various processes that severely deform the material
- The severe plastic deformation causes significant dislocation hardening, that can make the tube material too brittle
- Plastic deformation along slip planes results in reorientation of the grains
- The properties of the material are controlled by heat treatment
- Raising the temperature, but below 863 C, anneals the sample to reduce cold work: SRA
- Raising the temperature above 863 C changes to the phase. They then quench the sample to create a random texture in the phase: RXA or pRXA

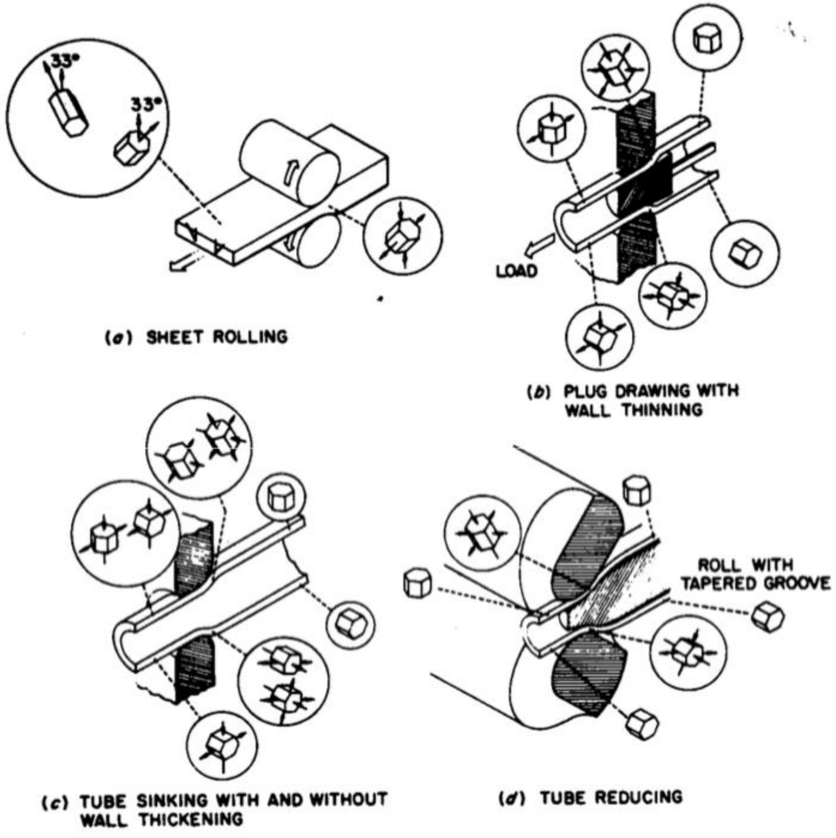


FIG. 48. Strain states for various types of fabrication of Zircaloy-2 [109].

Zr alloy microstructures

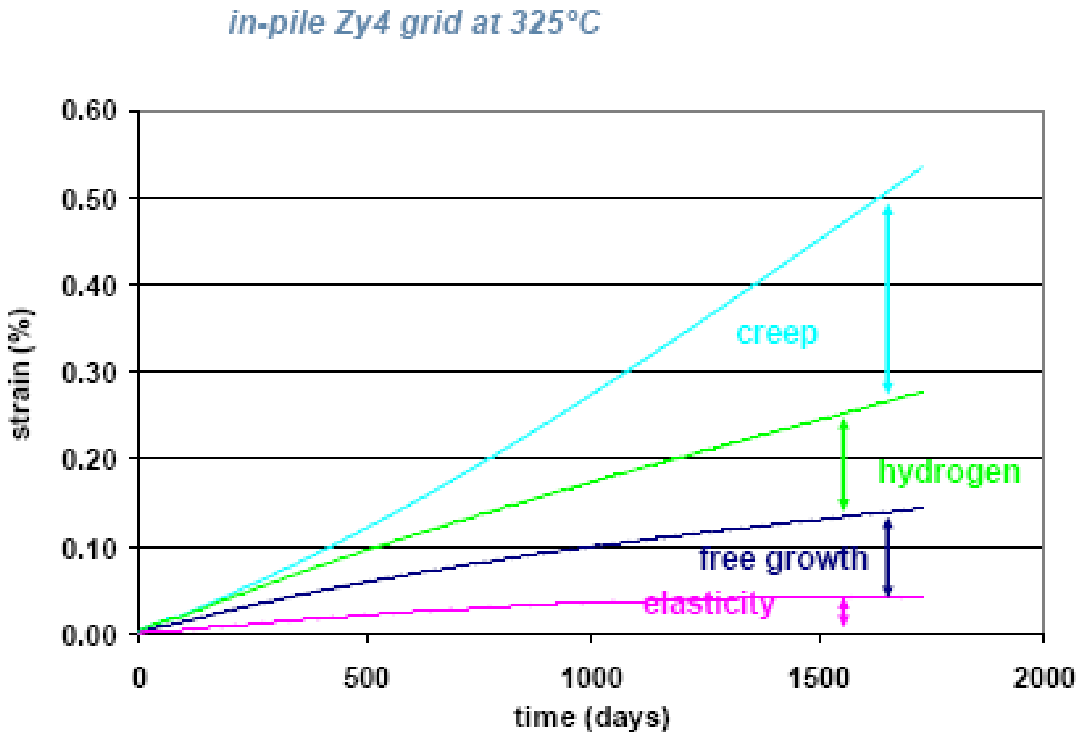
Fully recrystallized after quench

Stress-relieved microstructure

Zirconium Creep and Growth

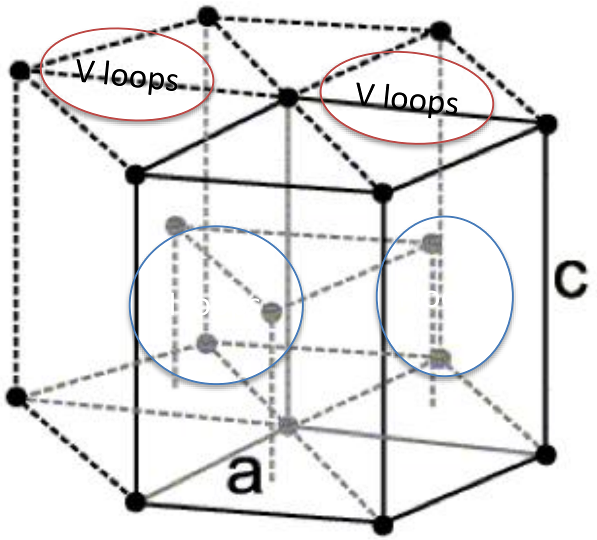
- The cladding undergoes a very large amount of irradiation during its life, the main effects are:
    - Interstitial loops form on prismatic planes
    - Later in the reactor exposure, vacancy loops form on the basal plane
- The overall number of displacements to the cladding can be calculated with the NRT model to be ~ 8 x 10^{-8} dpa/s, which, over 3 years exposure gives a total of ~8 dpa (every atom in the solid is displaced on the average eight times)

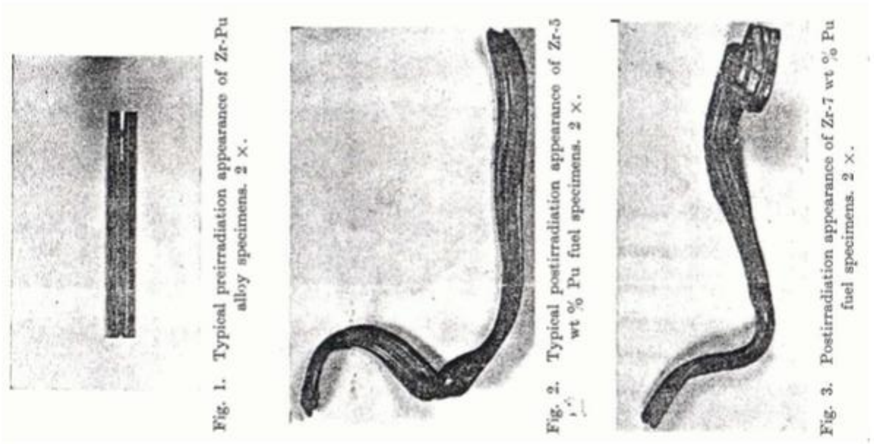
$$\nu(T) = \frac{\kappa E_D}{2E_d} = \frac{\kappa(T-\eta)}{2E_d},$$



Irradiation Growth

- Irradiation growth results from material anisotropy
- There is corresponding anisotropy in the defect behavior within the unit cell
- There is also a texture formed in the grain orientations
- Interstitial loops form on pyramidal planes, causing shrinkage along the center axis
- Later, vacancy loops form on basal planes, making it even worse
- The result is a change in shape, shrinking in the 0001 direction and expanding in the pyramidal direction





Irradiation Growth

- Three stages of irradiation growth
  - Initial rapid growth to small strains defect generation
  - Slow growth   defect accumulation, gets skipped in cold worked material
  - Accelerated or breakaway growth, caused by formation of vacancy loops on the basal plane
- Irradiation growth is affected by:
  - Fluence, Cold work, Texture, Irradiation temperature, Material chemistry, Hydrogen effects

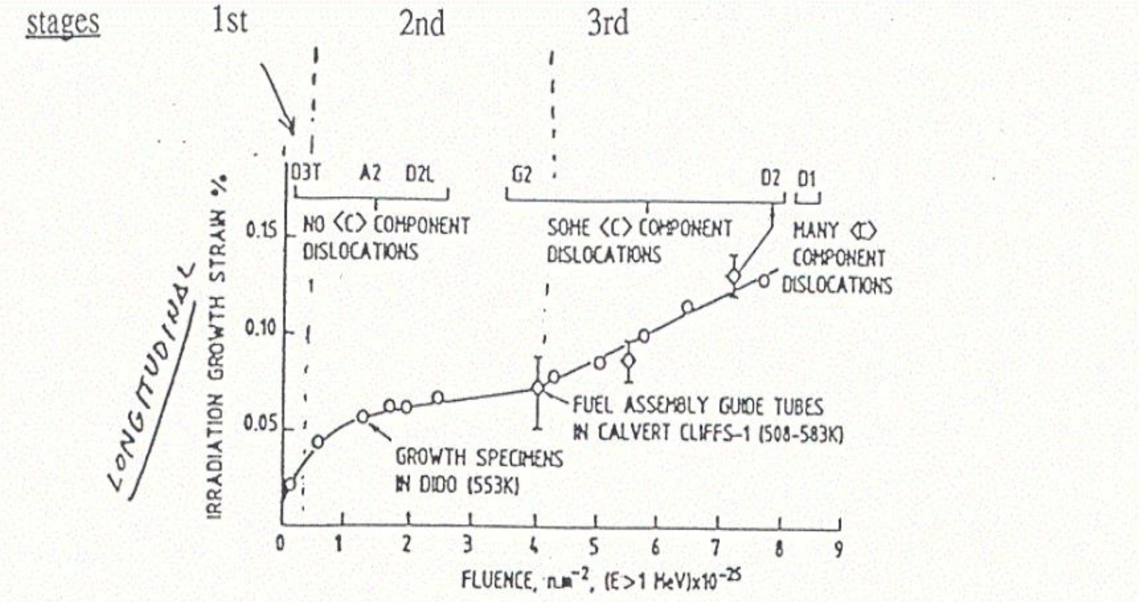
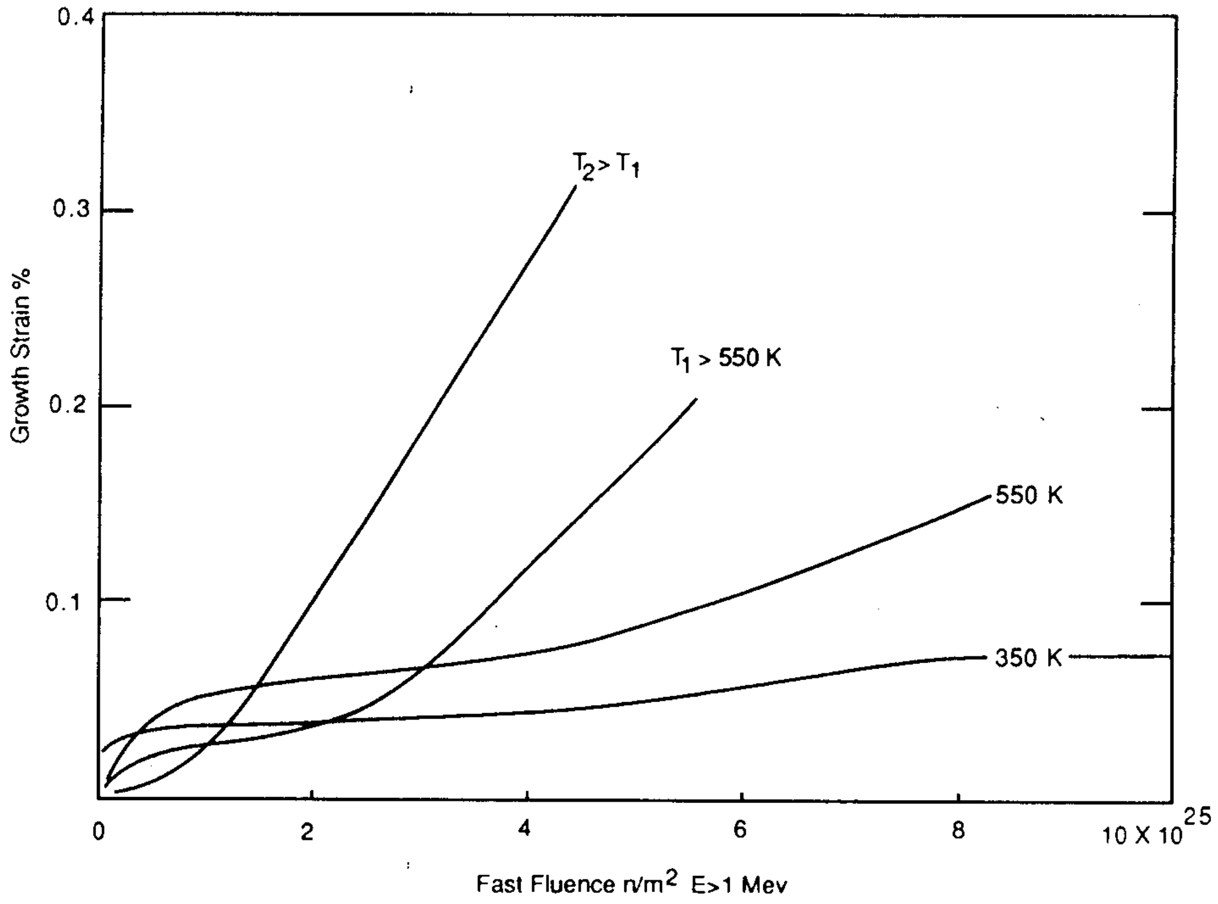


FIG. 11—*Irradiation growth in annealed Zircaloy at 550 to 580 K, showing accelerating growth at 4 ×* 10^{25} *n/m²*, E > *1 MeV* [61].

Irradiation Growth

- Higher temperature leads to earlier breakaway and higher growth
- Higher temperature adds mobility to defects, allowing them to diffuse faster, form clusters, and exacerbate the defect cluster induced anisotropic growth



Zircaloy Creep

- Empirical models have been developed for thermal and irradiation creep of Zircaloy
- Both based on the Von Mises stress

$$\sigma_m = \sqrt{\frac{1}{2}\left((\sigma_{11}-\sigma_{22})^2 + (\sigma_{22}-\sigma_{33})^2 + (\sigma_{22}-\sigma_{11})^2 + 6\left(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2\right)\right)}$$

- Thermal Creep

$$\dot{\epsilon}_{ss} = A_0\left(\frac{\sigma_m}{G}\right)^n e^{\left(\frac{-Q}{RT}\right)}$$

With A_0 = 3.14 × 10^{24} (1/s); shear modulus G = 4.2519 × 10^{10} 2.2185 × $10^7 T$ Pa; n = 5; Q = 2.7 × 10^5 J/mol

- Irradiation Creep

$$\dot{\epsilon}_{ir} = C_0 \Phi^{C_1} \sigma_m^{C_2}$$

φ is the fast neutron flux n/(cm^2 s) = 3E11xLHR n/(cm^2-s)

Note that SRA stands for stress relief annealed

RXA for recrystallization annealed

PRXA stands for partially recrystallization annealed

Clad Type	C_0	C_1	C_2
SRA	3.557×10^{-24}	0.85	1.0
RXA	1.654×10^{-24}	0.85	1.0
PRXA	2.714×10^{-24}	0.85	1.0
ZIRLO	2.846×10^{-24}	0.85	1.0

Zircaloy Creep Example

- Consider an SRA cladding tube at T = 600 K and LHR = 250 W/cm, with a stress $_m$ = 200 MPa. What is the total creep strain after three years?
- First, we will calculate the thermal creep

$$\dot{\epsilon}_{ss} = A_0\left(\frac{\sigma_m}{G}\right)^n e^{\left(\frac{-Q}{RT}\right)}$$

  $A_0 = 3.14 \times 10^{24}$ (1/s)

  G = 4.2519×10^{10} $2.2185 \times 10^{7} T$ Pa = 4.2519e10 2.2185e7*600 = 2.92e10 Pa

  Q = 2.7×10^{5} J/mol, n = 5, R = 8.3144598 J/(K mol)

  3.14e24*(200/2.92e4)^5*exp(-2.7e5/(8.3144598*600)) = 1.48e-10 1/s

- Now we will calculate the irradiation creep

$$\dot{\epsilon}_{ir} = C_0 \Phi^{C_1} {\sigma_m}^{C_2}$$

  C_0 = 3.557e-24, C_1 = 0.85, C_2 = 1.0

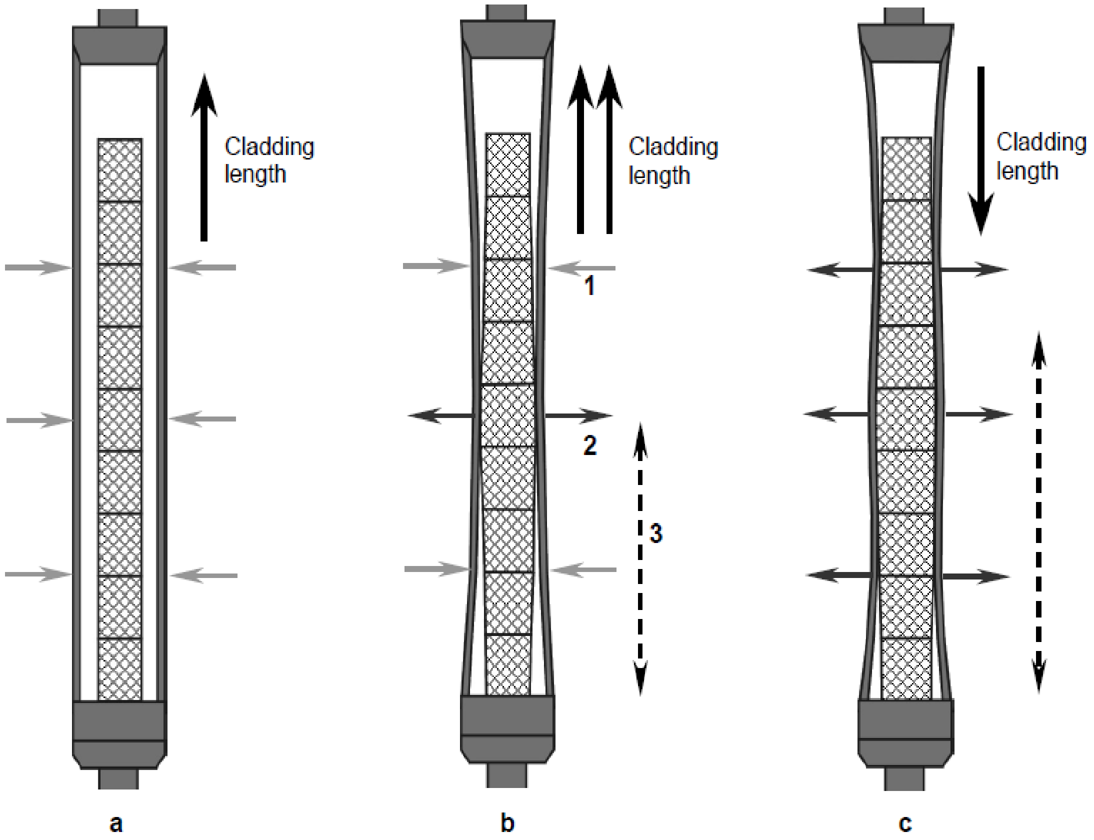
  ≈ 3e11 LHR = 3e11*250 = 7.5e13 n/(cm^2 s)

  3.557e-24*(7.5e13)^0.85*200^1 = 4.43e-10 1/s

- The total creep strain rate is 1.48e-10 + 4.43e-10 = 5.91e-10 1/s
- The total creep strain after three years is (assuming constant conditions) 5.91e-10*(3600*24*365*3) = 0.056 = 5.6% strain

Creep

- Creep impacts fuel performance by shrinking the gap and then conforming to the pellets
- a) before fuel-cladding interaction: The stress due to coolant pressure exceeds the internal stress from the gap; The diameter decreases due to thermal creep and irradiation creep, where the thermal creep decreases rapidly as irradiation damage builds up; The length increases due to anisotropic creep and irradiation growth



Creep

- b) start of fuel-cladding interaction: At the contact points the diameter increases, causing some contraction in rod length, and the expanding fuel imparts a local axial tensile stress on the rod, causing an increase in length; Irradiation growth causes an increase in rod length; The net change in length is the sum of the various inputs, but the net is an increase in rod length
- c) fuel-cladding interaction over most of fuel column: The fuel pellets stress the cladding outward, increasing the diameter of the rod; Anisotropic creep decreases the rod length; Axial pellet-cladding stresses and irradiation growth increase the rod length; The net change in rod length can become negative